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AGGREGATING SMALL REMOTE MEMORY ACCESS REQUESTS

Abstract

A network interface card (NIC) receives a stream of commands, a respective command comprising memory-operation requests, each request associated with a destination NIC. The NIC buffers asynchronously the requests into queues based on the destination NIC, each queue specific to a corresponding destination NIC. When first queue requests reach a threshold, the NIC aggregates the first queue requests into a first packet and sends the first packet to the destination NIC. The NIC receives a plurality of packets, a second packet comprising memory-operation requests, each request associated with a same destination NIC and a destination core. The NIC buffers asynchronously the requests of the second packet into queues based on the destination core, each queue specific to a corresponding destination core. When second queue requests reach the threshold, the NIC aggregates the second queue requests into a third packet and sends the third packet to the destination core.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/976,728, filed on Oct. 28, 2022. The disclosure of the aforementioned application is hereby incorporated by reference in its entirety.

BACKGROUND

Field

[0002] Current supercomputer performance may require a large number of small remote memory operations, where each operation may carry a certain network packet overhead. As link bandwidths increase, sustaining the line rate on such small packets can become more difficult and expensive. Systems which are based on commodity technologies may be under-achieving, while systems which are based on proprietary technologies may be too expensive.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0003] FIG. 1 illustrates a diagram of an architecture which facilitates aggregating remote memory-operation requests, in accordance with an aspect of the present application.

[0004] FIG. 2 illustrates an exemplary format of a NIC command including a header and multiple memory-operation requests, in accordance with an aspect of the present application.

[0005] FIG. 3 illustrates a diagram of a source NIC, including a first sort or asynchronous buffering of memory-operation requests based on a destination NIC, in accordance with an aspect of the present application.

[0006] FIG. 4 illustrates a diagram of a destination NIC, including a second sort or asynchronous buffering of memory-operation requests based on a destination core, in accordance with an aspect of the present application.

[0007] FIG. 5A presents a flowchart illustrating a method which facilitates aggregating remote memory-operation requests, including send operations by a single NIC, in accordance with an aspect of the present application.

[0008] FIG. 5B presents a flowchart illustrating a method which facilitates aggregating remote memory-operation requests, including receive operations by the single NIC of FIG. 5A, in accordance with an aspect of the present application.

[0009] FIG. 6A presents a flowchart illustrating a method which facilitates aggregating remote memory-operation requests, including send operations by a source NIC, in accordance with an aspect of the present application.

[0010] FIG. 6B presents a flowchart illustrating a method which facilitates aggregating remote memory-operation requests, including receive operations by a destination NIC, in accordance with an aspect of the present application.

[0011] FIG. 7 illustrates an apparatus which facilitates aggregating remote memory-operation requests, in accordance with an aspect of the present application.

[0012] In the figures, like reference numerals refer to the same figure elements.

DETAILED DESCRIPTION

[0013] The following description is presented to enable any person skilled in the art to make and use the aspects and examples, and is provided in the context of a particular application and its requirements. Various modifications to the disclosed aspects will be readily apparent to those skilled in the art, and the general principles defined herein may be applied to other aspects and applications without departing from the spirit and scope of the present disclosure. Thus, the aspects described herein are not limited to the aspects shown, but are to be accorded the widest scope consistent with the principles and features disclosed herein.

[0014] Current supercomputer performance may require a large number of small remote memory operations, where each operation may carry a certain network packet overhead. As link bandwidths increase, sustaining the line rate on such small packets can become more difficult and expensive. Systems which are based on commodity technologies may be under-achieving, while systems which are based on proprietary technologies may be too expensive. Furthermore, current solutions to provide efficient performance for small remote memory operations may involve aggregation of requests by software. However, aggregation of requests by software may consume large amounts of memory and central processing unit (CPU) time, which can lead to an inefficient use of those resources.

[0015] The described aspects of the present application address these challenges by providing a system which aggregates many small remote memory-operation requests into larger messages (e.g., a NIC command), where NICs (or NIC application-specific integrated circuits (ASICs)) can buffer the requests and process them asynchronously in order to service the requests in efficient batches. Instead of sending requests individually from a host to a source NIC, the requests can be streamed from the host to the source NIC. The source NIC can perform a first sort or a first asynchronous buffering (i.e., “buffered async” or “BA”), by placing each request into a source-side queue corresponding to a destination NIC indicated in the request. When a given source-side queue is full, the source NIC can send the data in the full queue as a packet to the destination NIC. The destination NIC can receive the packet and perform a second sort or a second asynchronous buffering by placing each request into a destination-side queue corresponding to a destination core indicated in the request. When a given destination-side queue is full, the destination NIC can send (i.e., stream) the data in the full queue as a packet to the destination core.

[0016] Thus, the described aspects can separate the process of sorting requests from the payload data carried by the requests, which can result in freeing up the software to provide the necessary resources based on the requirements of a given application. An exemplary high-level architecture with a source NIC and a destination NIC is described below in relation to FIG. 1, while detailed diagrams of a source (sending) NIC and a destination (receiving) NIC are described below in relation to, respectively, FIGS. 3 and 4.

[0017] The terms “buffering asynchronously,” “buffered async,” and “BA” are used interchangeably in this disclosure and refer to the operations described herein, in which requests are sorted or buffered into queues based on destination information, e.g.: based on the destination NIC in a first sort, as in request-sorting unit 124 and sorting engines 330 of, respectively, FIGS. 1 and 3); or based on the destination core in a second sort, as in request-sorting unit 152 and sorting engines 440 of, respectively, FIGS. 1 and 4). These requests are subsequently aggregated, e.g.: when based on the destination NIC, by data-sending unit 134 of FIG. 1 and request-aggregating unit 350 and packet-sending unit 352 of FIG. 3; and when based on the destination core, by data-streaming unit 142 of FIG. 1 and request-aggregating unit 424 and packet-sending unit 422 of FIG. 4.

[0018] The terms “endpoint” and “core” are used interchangeably in this disclosure and refer to one of a plurality endpoints or cores of a host associated with a given NIC.

[0019] The terms “memory-operation request,” “remote memory access request,” and “remote memory operation” are used interchangeably in this disclosure and refer to a request to access or perform an operation on a memory of a host. In this disclosure, these types of requests are

generally small in size. For example, 10-15 of these requests may fit into a single NIC command of 256 bytes. An exemplary NIC command with multiple memory-operation requests is depicted below in relation to FIG. 2.

Exemplary High-Level Architecture

[0020] FIG. 1 illustrates a diagram **100** of an architecture which facilitates aggregating remote memory-operation requests, in accordance with an aspect of the present application. Diagram **100** can include; a host with cores **110** (including, e.g., cores **112**, **114**, and **116**); a NIC **120**; a host with cores **160** (including, e.g., cores **162**, **164**, and **166**); and a NIC **140**. In an exemplary data path, NIC **120** (e.g., a source NIC) can send data to NIC **140** (e.g., destination NIC) over a network **102** (e.g., a high-bandwidth network). NIC **120** can include: a data-receiving unit **122**; a request-sorting unit **124**; a first plurality of queues **126**, **128**, **130**, and **132**; and a data-sending unit **134**. NIC **140** can include: a data-receiving unit **154**; a request-sorting unit **152**; a second plurality of queues **144**, **146**, **148**, and **150**; and a data-streaming unit **142**.

[0021] Queues **126-132** in the first plurality of queues in source NIC **120** can each be specific to a destination NIC. In some aspects, the number of queues in the first plurality of queues can be 4096, e.g., NIC **120** can communicate with up to 4096 other NICs. Queues **144-150** in the second plurality of queues in destination NIC **140** can each be specific to a destination core or endpoint. In some aspects, the number of queues in the second plurality of queues can be 256, e.g., NIC **140** can be associated with a host which has 256 cores or endpoints. The number of queues in the first plurality of queues of source NIC **120** and in the second plurality of queues of destination NIC **140** can be greater than or less than these exemplary values and can be based on various factors, e.g., specific application or customer needs, future changes to processor architecture or design, and variations in bandwidth.

[0022] Furthermore, based on current system designs, an exemplary size for each queue of both the first and second plurality of queues can be 256 bytes (not related to the number of actual queues of the destination NIC **140** corresponding to cores). As with the number of queues of both the first and second plurality of queues, the size of each queue (i.e., the depth of the queue) can be greater or less than this exemplary value and also based on various factors, as described above.

[0023] During operation, NIC **120**, by data-receiving unit **122**, can receive data from one of cores **110** (via, e.g., communications **168**, **169**, and **170**). Instead of a core sending data and the NIC receiving data as many individual small messages or requests, the core can stream data as NIC commands to data-receiving unit **122**. These streamed NIC commands can indicate that the payload is to be buffered and subsequently processed asynchronously (using the buffer async or BA method described here). A respective NIC command can further include a plurality of small remote memory-operation requests, each with a header and a payload. Each request can indicate a destination NIC and a destination endpoint or core. An exemplary NIC command is described below in relation to FIG. 2. Data-receiving unit **122** can transmit a received NIC command to request-sorting unit **124** (via a communication **172**). Request-sorting unit **124** can process the plurality of requests in the NIC command by asynchronously buffering the requests into a first plurality of queues (e.g., **126-132**) based on the destination NIC associated with or indicated in each request (via communications **174**). When a total size of the requests stored in a given queue reaches a predetermined threshold (e.g., 256 bytes), those requests can be aggregated into a first packet (via communications **176**) and data-sending unit **134** can send the first packet to the indicated destination NIC (via a communication **180** over network **102**).

[0024] NIC **140**, by data-receiving unit **154**, can receive a plurality of packets, including the first packet comprising the requests previously aggregated and stored in the given queue of NIC **120**. Recall that each request can indicate a destination NIC and a destination endpoint or core. Continuing with the example of the first packet received by data-receiving unit **154**, each of the requests in the first packet can indicate the same destination NIC and a destination endpoint or core associated with NIC **140**. Data-receiving unit **154** can transmit the received first packet to request-

sorting unit **152** (via a communication **184**). Request-sorting unit **152** can process the plurality of requests in the first packet by asynchronously buffering the requests into a second plurality of queues (e.g., **144-150**) based on the destination endpoint or core associated with or indicated in each request (via communications **186**). When a total size of the requests stored in a given queue reaches a predetermined threshold (e.g., 256 bytes), those requests can be aggregated into a packet (via communications **188**) and data-streaming unit **142** can send the packet to the indicated destination core (via, e.g., communications **190, 191, and 192**). Data-streaming unit **142** can thus stream packets destined for each specific core, where each packet is the size of the queue (e.g., 256 bytes) and contains many smaller remote memory-operation requests.

[0025] Thus, diagram **100** illustrates how the described aspects can reduce the overhead associated with transmitting a large number of memory-operation requests over a high-bandwidth network, by aggregating the requests into queues using a first sort on the source side (based on the destination NIC) and further using a second sort on the destination side (based on the destination core).

Exemplary Format of a NIC Command for Buffered Async

[0026] FIG. **2** illustrates an exemplary format of a NIC command **200** including a header and multiple memory-operation requests, in accordance with an aspect of the present application. NIC command **200** can be indicated by four-byte segments (e.g., a bytes 0-3 **210** segment, a bytes 4-7 **212** segment, a bytes 8-11 **214** segment, and a bytes 12-15 **216** segment). NIC command **200** can include as its first 16 bytes a BA command header **218** (indicated by a perpendicular cross-hatch fill pattern), which can indicate a length of its payload and that the payload comprises memory-operation requests which are to be buffered and aggregated asynchronously. For example, the length of the subsequent payload of command **200** can be 108 bytes.

[0027] The memory-operation requests in the payload of NIC command **200**

[0028] can each comprise a 4-byte header (indicated by a right-slanting fill pattern) and the corresponding payload. The 4-byte header can indicate at least the following: the destination NIC for the request; the destination core for the request; and a size or length of the payload. For example, a request 0 header **220** can indicate a destination NIC, a destination core, and a payload length of 8, while request 0 payload (bytes 0-7) **222** can follow. Subsequent requests can include a similar format: a request 1 header **224** can indicate its destination NIC, its destination core, and a payload length of 12 bytes, followed by a request 1 payload (bytes 0-11) **226**; a request 2 header **228** can indicate its destination NIC, its destination core, and a payload length of 56 bytes, followed by a request 2 payload (bytes 0-15) **230**, a request 2 payload (bytes 16-31) **232**, a request 2 payload (bytes 32-47) **234**, and a request 2 payload (bytes 48-55) **236**; and a request 3 header **238** can indicate its destination NIC, its destination core, and a payload length of 16 bytes, followed by a request 3 payload (bytes 0-3) **240** and a request 3 payload (bytes 4-15) **242**.

[0029] While NIC command **200** depicts only 16 bytes of BA command header **218** and 108 bytes of subsequent payload (for requests 0, 1, 2, and 3) for a total of 124 bytes, NIC command **200** can include data up to any predetermined value, e.g., 256 bytes. Memory-operation requests cannot be split across (i.e., do not cross) NIC command boundaries.

Detailed Description of Source/Sending NIC

[0030] FIG. **3** illustrates a diagram **300** of a source NIC **320**, including a first sort or asynchronous buffering of memory-operation requests based on a destination NIC, in accordance with an aspect of the present application. NIC **320** can receive data from cores **310** (e.g., one of cores **312, 314, 316, and 318** via communications **360**). NIC **320** can include: a data-receiving unit **322**; an engine-selecting unit **324**; a plurality of sorting engines **330** (e.g., eight engines **331, 332, 333, 334, 335, 336, 337, and 338**); a plurality of per destination-NIC queues **340** (e.g., queues **341, 342, 343, 344, 345, 346, and 347**); a request-aggregating unit **350**; and a packet-sending unit **352**.

[0031] As described above in relation to FIG. **1**, the cores can stream data as NIC commands to data-receiving unit **322**. These streamed NIC commands can indicate that the payload is to be buffered and subsequently processed asynchronously (using the buffer async or BA method

described here). Each NIC command can further include a plurality of small remote memory-operation requests, each with a header and a payload, as described above in relation to FIG. 2. Each request (in its header) can indicate a destination NIC and a destination endpoint or core.

[0032] Data-receiving unit **322** can transmit a received NIC command to engine-selecting unit **324** (via a communication **362**). Engine-selecting unit **324** can select, based on a load-balancing strategy, a first engine of a first plurality of engines (i.e., sorting engines **330**) and transmit a given NIC command to the selected engine (via a communication **364**). Each of the sorting engines can process a certain amount of data per clock cycle, e.g., 16 bytes per clock cycle. As a result, given a plurality (e.g., 8) of sorting engines, sorting engines **330** can buffer and process approximately $16 \times 8 = 128$ total bytes per clock cycle. A single NIC command will be processed entirely by the selected engine. That is, the NIC command will not be split further into smaller pieces for processing. Each sorting engine can process the requests in a given NIC command (e.g., at a rate of 16 bytes per clock cycle) and place the requests into the appropriate per-destination-NIC queue (of queues **340**).

[0033] For example, engine-selecting unit **324** may determine to send the given NIC command to sorting engine **334** (via a communication **366**). Sorting engine **334** can process the 256 bytes of the given NIC command by buffering each memory-operation request into the correct per destination-NIC queue (e.g., into queues **341-347** via communications **368** and **370**).

[0034] When a total size of the requests stored in a given queue of queues **340** reaches a predetermined threshold (e.g., 256 bytes), those requests can be aggregated into a first packet by request-aggregating unit **350** (via communications **378**). Request-aggregating unit **350** can send the first packet to packet-sending unit **352** (via a communication **380**), and packet-sending unit **352** can send the first packet to the indicated destination NIC (via a communication **382** over a network (not shown)).

Detailed Description of Destination/Receiving NIC

[0035] FIG. 4 illustrates a diagram **400** of a destination NIC **420**, including a second sort or asynchronous buffering of memory-operation requests based on a destination core, in accordance with an aspect of the present application. NIC **420** can include: a data-receiving unit **452**; an engine-selecting unit **450**; a plurality of sorting engines **440** (e.g., eight engines **441, 442, 443, 444, 445, 446, 447, and 448**); a plurality of per destination-core queues **430** (e.g., queues **431, 432, 433, 434, 4345, 436, and 437**); a request-aggregating unit **424**; and a packet-sending unit **422**. NIC **420** can stream data packets to cores/endpoints **410** (e.g., one of cores **412, 414, 416, and 418** via communications **476**).

[0036] NIC **420** can receive data from a source NIC (via a communication **460** over a network (not shown)). The data can include packets which comprise a plurality of small remote memory-operation requests, each with a header and a payload, as described above in relation to FIG. 2. Each request (in its header) can indicate a destination NIC and a destination endpoint or core. Packets received by destination NIC **420** can indicate the same destination NIC (i.e., NIC **420**) and a destination core (i.e., one of cores **410**). Data-receiving unit **452** can transmit a received packet to engine-selecting unit **450** (via a communication **462**). Engine-selecting unit **450** can select, based on a load-balancing strategy, a second engine of a second plurality of engines (i.e., sorting engines **440**) and transmit a given packet to the selected engine (via a communication **464**). Similar to sorting engines **330** depicted in FIG. 3 in source NIC **320**, each of the sorting engines **440** can process a certain amount of data per clock cycle, e.g., 16 bytes per clock cycle. As a result, given a plurality (e.g., 8) of sorting engines, sorting engines **330** can buffer and process approximately $16 \times 8 = 128$ total bytes per clock cycle. A single packet will be processed entirely by the selected engine. That is, the packet will not be split further into smaller pieces for processing. Each sorting engine can process the requests in a given packet (e.g., at a rate of 16 bytes per clock cycle) and place the requests into the appropriate per destination-core queue (of queues **430**).

[0037] For example, engine-selecting unit **450** may determine to send the given packet to sorting

engine **445** (via a communication **466**). Sorting engine **445** can process the (up to) 256 bytes of the given packet by buffering each memory-operation request into the correct per destination-core queue (e.g., into queues **431-437** via communications **468** and **470**).

[0038] When a total size of the requests stored in a given queue of queues **430** reaches a predetermined threshold (e.g., 256 bytes), those requests can be aggregated into a second packet by request-aggregating unit **424** (via communications **472**). Request-aggregating unit **424** can send the second packet to packet-sending unit **422** (via a communication **474**), and packet-sending unit **422** can send the second packet to the indicated destination core (via communications **476**). Packet-sending unit **422** can be a data-streaming unit, i.e., stream multiple packets destined to the various cores of destination NIC **420**.

Method for Facilitating Aggregation of Remote Memory-Operation Requests

[0039] The described aspects can include a single NIC which performs both source (send) operations and destination (receive) operations, as depicted below in relation to FIGS. 5A and 5B. FIG. 5A presents a flowchart **500** illustrating a method which facilitates aggregating remote memory-operation requests, including send operations by a single NIC, in accordance with an aspect of the present application. During operation, the system receives, by a local network interface card (NIC), a stream of commands, wherein a respective command comprises a first plurality of memory-operation requests, wherein each request is associated with a remote destination NIC and a remote destination core (operation **502**). The stream of commands is received by the local NIC as a stream rather than as individual memory-operation requests. The local NIC can retrieve data in contiguous arrays of memory-operation requests comprising payload and corresponding destination information (destination NIC and destination core) over, e.g., a Peripheral Component Interconnect Express (PCIe) connection, a Compute Express Link (CXL), or other host interface or on-chip network. The system buffers asynchronously the requests into a first plurality of queues based on the destination NIC associated with each request, wherein each queue is specific to a corresponding remote destination NIC (operation **504**). If a total size of requests stored in a first queue does not reach a predetermined threshold (decision **506**), the system can continue at operation **502** or **504**.

[0040] If a total size of requests stored in a first queue does reach a predetermined threshold (decision **506**), the system aggregates the requests stored in the first queue into a first packet (operation **508**) and sends the first packet to the remote destination NIC over a high-bandwidth network, thereby reducing overhead associated with transmitting a large number of memory-operation requests over the high-bandwidth network (operation **510**). The operation continues at Label A of FIG. 5B.

[0041] FIG. 5B presents a flowchart **520** illustrating a method which facilitates aggregating remote memory-operation requests, including receive operations by the single NIC of FIG. 5A, in accordance with an aspect of the present application. The system receives, by the local NIC, a plurality of packets, wherein a second packet comprises a second plurality of memory-operation requests, wherein each request is destined to the local NIC and associated with a local destination core (operation **522**). The system buffers asynchronously the requests of the second packet into a second plurality of queues based on the destination core associated with each request, wherein each queue is specific to a corresponding local destination core (operation **524**). If a total size of the requests stored in a second queue of the second plurality of queues does not reach the predetermined threshold (decision **526**), the system can continue at operation **522** or **524**.

[0042] If a total size of the requests stored in a second queue of the second plurality of queues does reach the predetermined threshold (decision **526**), the system aggregates the requests stored in the second queue into a third packet (decision **528**) and sends the third packet to the local destination core, thereby further reducing the overhead associated with transmitting a large number of memory-operation requests over the high-bandwidth network (operation **530**). The system, by the local NIC, can determine that a total size of aggregated requests stored in one or more queues of

the second plurality of queues reaches the predetermined threshold and can further stream those aggregated requests to a respective corresponding local destination core specific to a respective queue (not shown). The operation returns.

[0043] The described aspects can also include a system which includes two NICs: a first NIC (e.g., a source NIC or a sending NIC); and a second NIC (e.g., a destination NIC or a receiving NIC), as depicted below in relation to FIGS. 6A and 6B. FIG. 6A presents a flowchart **600** illustrating a method which facilitates aggregating remote memory-operation requests, including send operations by a source NIC, in accordance with an aspect of the present application. During operation, the system receives, by a source network interface card (NIC), a stream of commands, wherein a respective command comprises a first plurality of memory-operation requests, wherein each request is associated with a destination NIC and a destination core (operation **602**). The system buffers asynchronously the requests into a first plurality of queues based on the destination NIC associated with each request, wherein each queue is specific to a corresponding destination NIC (operation **604**). If a total size of requests stored in a first queue does not reach a predetermined threshold (decision **606**), the system can continue at operation **602** or **604**.

[0044] If a total size of requests stored in a first queue does reach a predetermined threshold (decision **606**), the system aggregates the requests stored in the first queue into a first packet (operation **608**) and sends the first packet to the destination NIC over the high-bandwidth network, thereby reducing overhead associated with transmitting a large number of memory-operation requests over the high-bandwidth network (operation **610**). The operation continues at Label B of FIG. 6B.

[0045] FIG. 6B presents a flowchart **620** illustrating a method which facilitates aggregating remote memory-operation requests, including receive operations by a destination NIC, in accordance with an aspect of the present application. The system receives, by a destination NIC, a plurality of packets, including the first packet comprising the requests previously aggregated and stored in the first queue of the first plurality of queues, wherein each request is associated with a same destination NIC and a destination core (operation **622**). The system, by the destination NIC, buffers asynchronously the requests of the first packet into a second plurality of queues based on the destination core associated with each request, wherein each queue is specific to a corresponding destination core (operation **624**). If a total size of the requests stored in a second queue of the second plurality of queues does not reach the predetermined threshold (decision **626**), the system can continue at operation **622** or **624**.

[0046] If a total size of the requests stored in a second queue of the second plurality of queues does not reach the predetermined threshold (decision **626**), the system, by the destination NIC, aggregates the requests stored in the second queue into a second packet (operation **628**) and sends the second packet to the destination core, thereby reducing overhead associated with transmitting a large number of memory-operation requests over the high-bandwidth network (operation **630**). The operation returns.

Apparatus Which Facilitates Aggregating Remote Memory-Operation Requests

[0047] FIG. 7 illustrates an apparatus **700** which facilitates aggregating remote memory-operation requests, in accordance with an aspect of the present application. Apparatus **700** can represent a network interface card (NIC) (such as the single NIC described in relation to flowcharts **500** and **520** of FIGS. 5A and 5B) and can include a send unit **710** and a receive unit **720**. Send unit **710** can include: a first command unit **712** (which can perform operations similar to those described above in relation to data-receiving units **122** and **322** of, respectively, FIGS. 1 and 3); a first sorting unit **714** (which can perform operations similar to those described above in relation to request-sorting unit **124** and sorting engines **330** of, respectively, FIGS. 1 and 3); a first queue-managing unit **716** (which can manage and buffer data in queues **126-132** of FIG. 1 and in per destination-NIC queues **340** of FIG. 3); and a first aggregation-communication unit **718** (which can perform operations similar to those described above for data-sending unit **134** of FIG. 1 and request-aggregating unit

350 and packet-sending unit **352** of FIG. 3).

[0048] Receive unit **720** can include: a second command unit **722** (which can perform operations similar to those described above in relation to data-receiving units **154** and **452** of, respectively, FIGS. **1** and **4**); a second sorting unit **724** (which can perform operations similar to those described above in relation to request-sorting unit **152** and sorting engines **440** of, respectively, FIGS. **1** and **4**); a second queue-managing unit **726** (which can manage and buffer data in queues **144-150** of FIG. **1** and in per destination-core queues **430** of FIG. **4**); and a second aggregation-communication unit **728** (which can perform operations similar to those described above for data-streaming unit **142** of FIG. **1** and request-aggregating unit **424** and packet-sending unit **422** of FIG. **4**).

[0049] In general, the disclosed aspects provide a system, method, and apparatus which facilitates. In one aspect, the system receives, by a local network interface card (NIC), a stream of commands, wherein a respective command comprises a first plurality of memory-operation requests, wherein each request is associated with a remote destination NIC and a remote destination core. The system buffers asynchronously the requests into a first plurality of queues based on the destination NIC associated with each request, wherein each queue is specific to a corresponding remote destination NIC. Responsive to determining that a total size of the requests stored in a first queue reaches a predetermined threshold, the system aggregates the requests stored in the first queue into a first packet and sends the first packet to the remote destination NIC over a high-bandwidth network. The system receives, by the local NIC, a plurality of packets, wherein a second packet of the received packets comprises a second plurality of memory-operation requests, wherein each request is destined to the local NIC and associated with a local destination core. The system buffers asynchronously the requests of the second packet into a second plurality of queues based on the local destination core associated with each request, wherein each queue is specific to a corresponding local destination core. Responsive to determining that a total size of the requests stored in a second queue of the second plurality of queues reaches the predetermined threshold, the system aggregates the requests stored in the second queue into a third packet and sends the third packet to the local destination core.

[0050] In a variation on this aspect, a first plurality of engines of the local NIC buffers asynchronously the requests into the first plurality of queues, and the system selects, based on a load-balancing strategy, a first engine of the first plurality of engines to buffer asynchronously the requests from each command.

[0051] In a further variation on this aspect, a second plurality of engines of the local NIC buffers asynchronously the requests into the second plurality of queues, and the system selects, based on a load-balancing strategy, a second engine of the second plurality of engines to buffer asynchronously the requests from each packet.

[0052] In a further variation, the stream of commands is received by the local NIC as a stream of commands and not as individual memory-operation requests.

[0053] In a further variation, the local NIC receives the stream of commands by retrieving data in contiguous arrays of memory-operation requests comprising payloads and corresponding destination information over a Peripheral Component Interconnect Express (PCIe) connection.

[0054] In a further variation, the system determines that a total size of aggregated requests stored in one or more queues of the second plurality of queues reaches the predetermined threshold. The system streams, by the local NIC, the aggregated requests stored in the one or more queues of the second plurality of queues to a respective corresponding destination core specific to a respective queue.

[0055] In a further variation, a respective remote destination core corresponds to a destination endpoint of a plurality of destination endpoints associated with the remote destination NIC.

[0056] In a further variation, a respective memory-operation request is associated with a payload of a size smaller than a predetermined size.

[0057] In a further variation, each command received by the local NIC is of a size up to 256 bytes.

[0058] In a further variation, the first plurality of queues comprises 4096 queues, and the second plurality of queues comprises 256 queues.

[0059] In a further variation, the respective command indicates in a header that the memory-operation requests are to be buffered and aggregated asynchronously.

[0060] In another aspect, an apparatus or a NIC comprises: a first command module to receive a stream of commands, wherein a respective command comprises a first plurality of memory-operation requests, wherein each request is associated with a remote destination NIC and a remote destination core; a first sorting module to buffer asynchronously the requests into a first plurality of queues based on the remote destination NIC associated with each request, wherein each queue is specific to a corresponding remote destination NIC; a first aggregation-communication module to, responsive to determining that a total size of the requests stored in a first queue of the first plurality of queues reaches a predetermined threshold, aggregate the requests stored in the first queue into a first packet and send the first packet to the remote destination NIC over a high-bandwidth network; a second command module to receive a plurality of packets, wherein a second packet of the received packets comprises a second plurality of memory-operation requests, wherein each request is destined to the NIC and associated with a local destination core; a second sorting module to buffer asynchronously the requests of the second packet into a second plurality of queues based on the local destination core associated with each request, wherein each queue is specific to a corresponding local destination core; and a second aggregation-communication module to, responsive to determining that a total size of the requests stored in a second queue of the second plurality of queues reaches the predetermined threshold, aggregate the requests stored in the second queue into a third packet and send the third packet to the local destination core.

[0061] In yet another aspect, a system comprises a local NIC (e.g., a source NIC) and a remote NIC (e.g., a destination NIC). The local NIC comprises: a first command module to receive a stream of commands, wherein a respective command comprises a first plurality of memory-operation requests, wherein each request is associated with a remote destination NIC and a remote destination core; a first sorting module to buffer asynchronously the requests into a first plurality of queues based on the destination NIC associated with each request, wherein each queue is specific to a corresponding remote destination NIC; and a first aggregation-communication module to, responsive to determining that a total size of the requests stored in a first queue of the first plurality of queues reaches a predetermined threshold, aggregate the requests stored in the first queue into a first packet and send the first packet to the remote destination NIC over a high-bandwidth network. The remote NIC comprises: a second command module to receive the first packet comprising the requests previously aggregated and stored in the first queue, wherein each request is destined to the remote NIC and associated with a remote destination core; a second sorting module to buffer asynchronously the requests of the first packet into a second plurality of queues based on the remote destination core associated with each request, wherein each queue is specific to a corresponding remote destination core; and a second aggregation-communication module to, responsive to determining that a total size of the requests stored in a second queue of the second plurality of queues reaches the predetermined threshold, aggregate the requests stored in the second queue into a second packet and send the second packet to the remote destination core.

[0062] The foregoing descriptions of aspects have been presented for purposes of illustration and description only. They are not intended to be exhaustive or to limit the aspects described herein to the forms disclosed. Accordingly, many modifications and variations will be apparent to practitioners skilled in the art. Additionally, the above disclosure is not intended to limit the aspects described herein. The scope of the aspects described herein is defined by the appended claims.

Claims

- 1.** A computer-implemented method comprising: receiving, by a local network interface card (NIC), a stream of commands, wherein a respective command comprises a first plurality of memory-operation requests, wherein each request is associated with a remote destination NIC and a remote destination core; buffering asynchronously the requests into a first plurality of queues based on the destination NIC associated with each request, wherein each queue is specific to a corresponding remote destination NIC; responsive to determining that a total size of the requests stored in a first queue reaches a predetermined threshold, aggregating the requests stored in the first queue into a first packet and sending the first packet to the remote destination NIC over a high-bandwidth network; receiving, by the local NIC, a plurality of packets, wherein a second packet of the received packets comprises a second plurality of memory-operation requests, wherein each request is destined to the local NIC and associated with a local destination core; buffering asynchronously the requests of the second packet into a second plurality of queues based on the local destination core associated with each request, wherein each queue is specific to a corresponding local destination core; and responsive to determining that a total size of the requests stored in a second queue of the second plurality of queues reaches the predetermined threshold, aggregating the requests stored in the second queue into a third packet and sending the third packet to the local destination core.
- 2.** The method of claim 1, wherein a first plurality of engines of the local NIC buffers asynchronously the requests into the first plurality of queues, and wherein the method further comprises selecting, based on a load-balancing strategy, a first engine of the first plurality of engines to buffer asynchronously the requests from each command.
- 3.** The method of claim 1, wherein a second plurality of engines of the local NIC buffers asynchronously the requests into the second plurality of queues, and wherein the method further comprises selecting, based on a load-balancing strategy, a second engine of the second plurality of engines to buffer asynchronously the requests from each packet.
- 4.** The method of claim 1, wherein the stream of commands is received by the local NIC as a stream of commands and not as individual memory-operation requests.
- 5.** The method of claim 1, further comprising: wherein the local NIC receives the stream of commands by retrieving data in contiguous arrays of memory-operation requests comprising payloads and corresponding destination information over a Peripheral Component Interconnect Express (PCIe) connection.
- 6.** The method of claim 1, further comprising: determining that a total size of aggregated requests stored in one or more queues of the second plurality of queues reaches the predetermined threshold; and streaming, by the local NIC, the aggregated requests stored in the one or more queues of the second plurality of queues to a respective corresponding destination core specific to a respective queue.
- 7.** The method of claim 1, wherein a respective remote destination core corresponds to a destination endpoint of a plurality of destination endpoints associated with the remote destination NIC.
- 8.** The method of claim 1, wherein a respective memory-operation request is associated with a payload of a size smaller than a predetermined size.
- 9.** The method of claim 1, wherein each command received by the local NIC is of a size up to 256 bytes.
- 10.** The method of claim 1, wherein the first plurality of queues comprises 4096 queues, and wherein the second plurality of queues comprises 256 queues.
- 11.** The method of claim 1, wherein the respective command indicates in a header that the memory-operation requests are to be buffered and aggregated asynchronously.
- 12.** A network interface card (NIC) comprising: a first command module to receive a stream of commands, wherein a respective command comprises a first plurality of memory-operation requests, wherein each request is associated with a remote destination NIC and a remote destination

core; a first sorting module to buffer asynchronously the requests into a first plurality of queues based on the remote destination NIC associated with each request, wherein each queue is specific to a corresponding remote destination NIC; a first aggregation-communication module to, responsive to determining that a total size of the requests stored in a first queue of the first plurality of queues reaches a predetermined threshold, aggregate the requests stored in the first queue into a first packet and send the first packet to the remote destination NIC over a high-bandwidth 12 network; a second command module to receive a plurality of packets, wherein a second packet of the received packets comprises a second plurality of memory-operation requests, wherein each request is destined to the NIC and associated with a local destination core; a second sorting module to buffer asynchronously the requests of the second packet into a second plurality of queues based on the local destination core associated with each request, wherein each queue is specific to a corresponding local destination core; and a second aggregation-communication module to, responsive to determining that a total size of the requests stored in a second queue of the second plurality of queues reaches the predetermined threshold, aggregate the requests stored in the second queue into a third packet and send the third packet to the local destination core.

13. The NIC of claim 12, further comprising: a first queue-managing module to manage and store requests buffered by the first sorting module into the first plurality of queues; and a second queue-managing module to manage and store requests buffered by the second sorting module into the second plurality of queues.

14. The NIC of claim 12, wherein the first sorting module comprises a first plurality of engines which buffer asynchronously the requests into the first plurality of queues, and wherein the NIC further comprises a first engine-selecting module to select, based on a load-balancing strategy, a first engine of the first plurality of engines to buffer asynchronously the requests from each command.

15. The NIC of claim 12, wherein the second sorting module comprises a second plurality of engines which buffer asynchronously the requests into the second plurality of queues, and wherein the NIC further comprises a second engine-selecting module to select, based on a load-balancing strategy, a second engine of the second plurality of engines to buffer asynchronously the requests from each packet.

16. The NIC of claim 12, wherein the first command module is further to receive the stream of commands by retrieving data in contiguous arrays of memory-operation requests comprising payloads and corresponding destination information over a Peripheral Component Interconnect Express (PCIe) connection.

17. The NIC of claim 12, wherein a respective queue in the second plurality of queues corresponds to one of a plurality of local destination cores or destination endpoints associated with the first NIC.

18. The NIC of claim 12, wherein a respective memory-operation request is associated with a payload of a size smaller than a predetermined size, wherein each command received by the first command module and each packet received by the second command module is of a size up to 256 bytes, wherein the first plurality of queues comprises 4096 queues, and wherein the second plurality of queues comprises 256 queues.

19. A system comprising: a local network interface card (NIC), comprising: a first command module to receive a stream of commands, wherein a respective command comprises a first plurality of memory-operation requests, wherein each request is associated with a remote destination NIC and a remote destination core; a first sorting module to buffer asynchronously the requests into a first plurality of queues based on the destination NIC associated with each request, wherein each queue is specific to a corresponding remote destination NIC; and a first aggregation-communication module to, responsive to determining that a total size of the requests stored in a first queue of the first plurality of queues reaches a predetermined threshold, aggregate the requests stored in the first queue into a first packet and send the first packet to the remote destination NIC

over a high-bandwidth network; and a remote NIC, comprising: a second command module to receive the first packet comprising the requests previously aggregated and stored in the first queue, wherein each request is destined to the remote NIC and associated with a remote destination core; a second sorting module to buffer asynchronously the requests of the first packet into a second plurality of queues based on the remote destination core associated with each request, wherein each queue is specific to a corresponding remote destination core; and a second aggregation-communication module to, responsive to determining that a total size of the requests stored in a second queue of the second plurality of queues reaches the predetermined threshold, aggregate the requests stored in the second queue into a second packet and send the second packet to the remote destination core.

20. The system of claim 19, wherein the local NIC further comprises a first queue-managing module to manage and store requests buffered by the first sorting module into the first plurality of queues, wherein the remote NIC further comprises a second queue-managing module to manage and store requests buffered by the second sorting module into the second plurality of queues, wherein the first sorting module of the local NIC comprises a first plurality of engines which buffer asynchronously the requests into the first plurality of queues, wherein the local NIC further comprises a first engine-selecting module to select, based on a load-balancing strategy, a first engine of the first plurality of engines to buffer asynchronously the requests from each command, wherein the second sorting module of the remote NIC comprises a second plurality of engines which buffer asynchronously the requests into the second plurality of queues, and wherein the remote NIC further comprises a second engine-selecting module to select, based on a load-balancing strategy, a second engine of the second plurality of engines to buffer asynchronously the requests from each packet.
